Pan/Tilt Redesign — Option B

Same small size you approved. This time, it still looks and works like the original.

Prepared for: a non-engineer reader | Prepared by: Mechanical Design Engineering | Report date: 2026-08-11

In one sentence

This document designs a second, alternative small pan/tilt build — call it **Option B** — that ends up roughly the same pocket-sized footprint as the redesign you already saw (built around the same small motor), but keeps far more of the original mechanism's actual parts, shapes, and character. The first attempt hit the size target by simplifying aggressively: it fused two parts into one, swapped every bearing for a generic type, and deleted four small parts entirely. You liked the size — you didn't like how much else changed. Option B fixes that: same ballpark size, but it stays close to the original's part-by-part structure, and only changes what the real motor or real hardware sizes genuinely force.

This report does not include a new 3D model file — it's a design study, in writing, with diagrams and a full parts list, in the same spirit as the earlier redesign document. A companion CAD file can be produced once you confirm this direction.

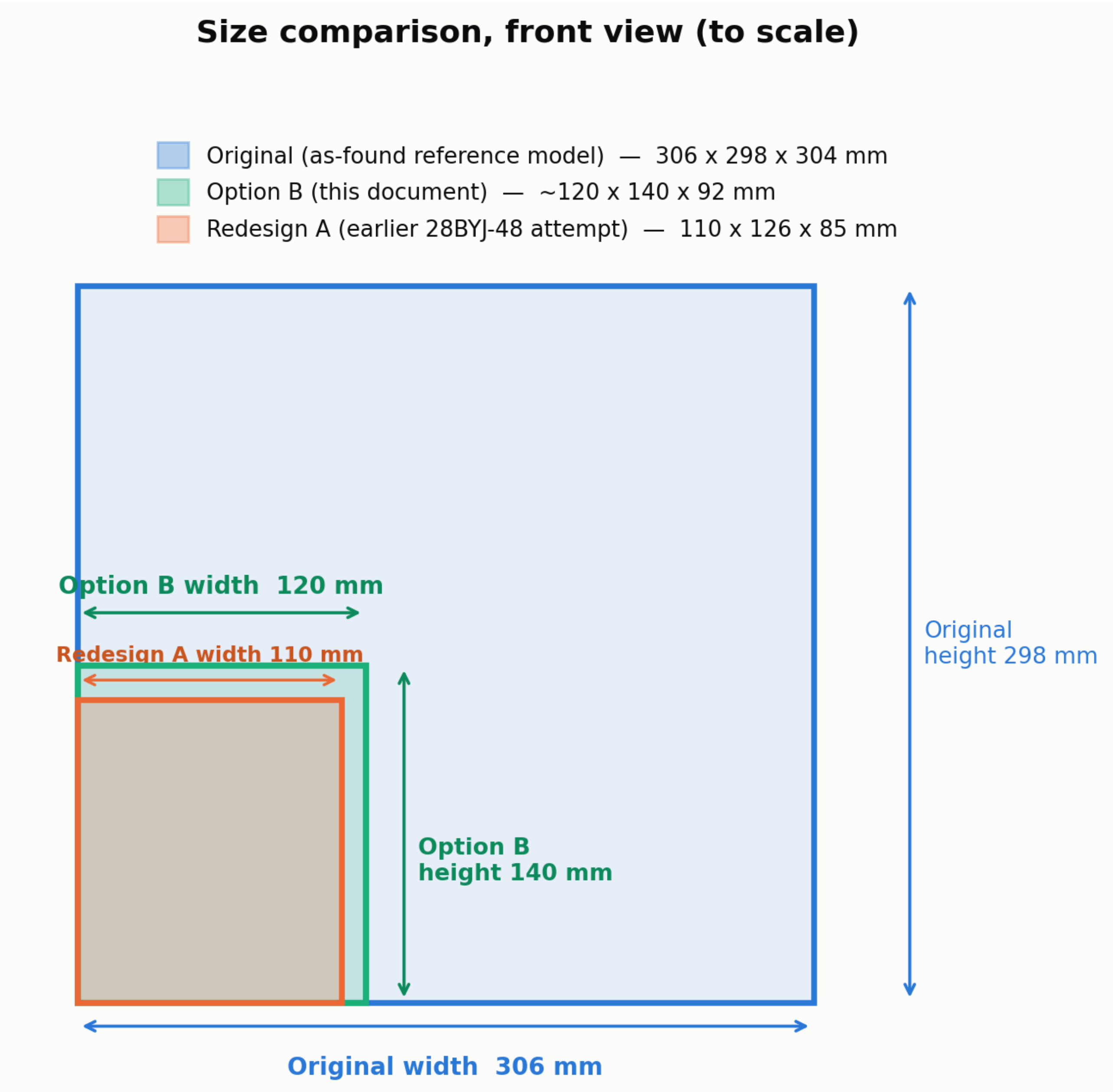


Figure 1 — Three-way size comparison, front view, drawn to scale. Option B sits between the original and the first redesign attempt: dramatically smaller than the original, and only a little larger than the first attempt — the extra room is spent on keeping parts separate rather than fusing them together.

1. Why this document exists

The starting point for all of this is a reference 3D file called **ASSEMBLY_pan_tilt.stp** — a fairly large design, about the size of a shoebox (306 x 298 x 304 mm), with two motorised swivel axes (a “pan” axis that spins side-to-side, and a “tilt” axis that tips up and down), 17 different part designs, and two placeholder motors that were never modelled as any specific real product.

A first attempt to shrink this down and build it around a real, cheap, widely-available motor (the **28BYJ-48** stepper motor) produced a design about the size of a fist — 110 x 126 x 85 mm. That's the right size. But getting there meant simplifying hard: it merged a bracket called “Cam Holder” directly into another part instead of keeping it separate, it replaced every original bearing with a generic, smaller substitute, it deleted four small spacer parts and turned them into printed bumps on other parts, and it cut the number of distinct part designs from 17 down to 11.

Your feedback on that attempt was clear: *“it doesn't look like the original anymore”* and *“it changed too much, not just the size.”* **Option B is a second attempt at the same size, that takes that feedback seriously.**

2. The rule Option B follows

> **The design rule**
>
> **Shrink it. Fix what's actually broken. Don't redesign it.**
>
> A change is only allowed into Option B if it's forced by one of two things: (a) the real motor's shape — its shaft doesn't line up with the middle of its body, so anything that touches the motor has to be redrawn — or (b) a part that's mathematically too small to exist at this scale (an original bearing built for a 25 mm-thick shaft simply isn't sold small enough for an 8 mm shaft). Everything else — what a part is called, how many separate parts there are, whether a bracket bolts on or gets welded into its neighbour — stays as close to the original as the smaller size allows.

This rule is why Option B ends up a little larger than the first attempt (roughly 120 x 140 x 92 mm instead of 110 x 126 x 85 mm) — keeping parts separate means each joint needs a little wall thickness and a couple of screw bosses that a single fused part wouldn't need. That's a small, deliberate trade: about 9% bigger in each direction, in exchange for a design that still visibly resembles where it came from.

3. What the new motor forces, before anything else

Before talking about what Option B kept, it helps to understand the one thing neither design had a choice about: the motor. Both the first redesign attempt and Option B are built around the same real, cheap, widely-used small motor — the **28BYJ-48**. A few basic facts about this motor decide a surprising amount of the design, and they apply equally to both versions:

It's geared down a lot inside its own case (about 64 turns of the internal motor for 1 turn of the output shaft).	The motor is naturally strong for its size and naturally slow — good for careful aiming, not for fast tracking.
It can push with at least 34.3 “milli-newton-metres” of turning force at its output shaft (think of this as a unit for “turning strength,” the rotational version of how hard you'd have to push on a lever).	Multiplied by the external gears (see below), this is more than enough to hold and move a small 15-gram payload like a little camera or sensor, with room to spare.
Its output shaft isn't in the middle of its round metal case — it's offset to one side, and it has two small flat sides ground onto it (like a two-sided key) instead of being perfectly round.	Any bracket that holds this motor has to be drawn specifically around it — you can't just shrink the original's round “motor-shaped hole” and expect it to fit.

Because of the offset shaft, **both** designs had to throw away the original's generic, placeholder motor mounts and redraw them from scratch around the real motor. This is not a stylistic choice — it's the one part of this whole project that is simply unavoidable no matter how much you want to preserve the original's look. Option B keeps the same practical answer the first attempt found: mount the pan motor with its shaft pointing straight down into the base, and mount the tilt motor tucked in behind the tilt gear rather than underneath it — both choices keep the design compact without adding unnecessary height.

The gears also keep the ratio the original used (each axis gears the motor down another 3-to-1, on top of the motor's own internal gearing) — but the size of each individual gear tooth has to shrink slightly, for a printing reason explained in section 5.

4. What Option B kept that the first attempt gave up

This is the heart of Option B. Four things the first redesign simplified away are put back, each adapted to the smaller size but kept recognisably itself.

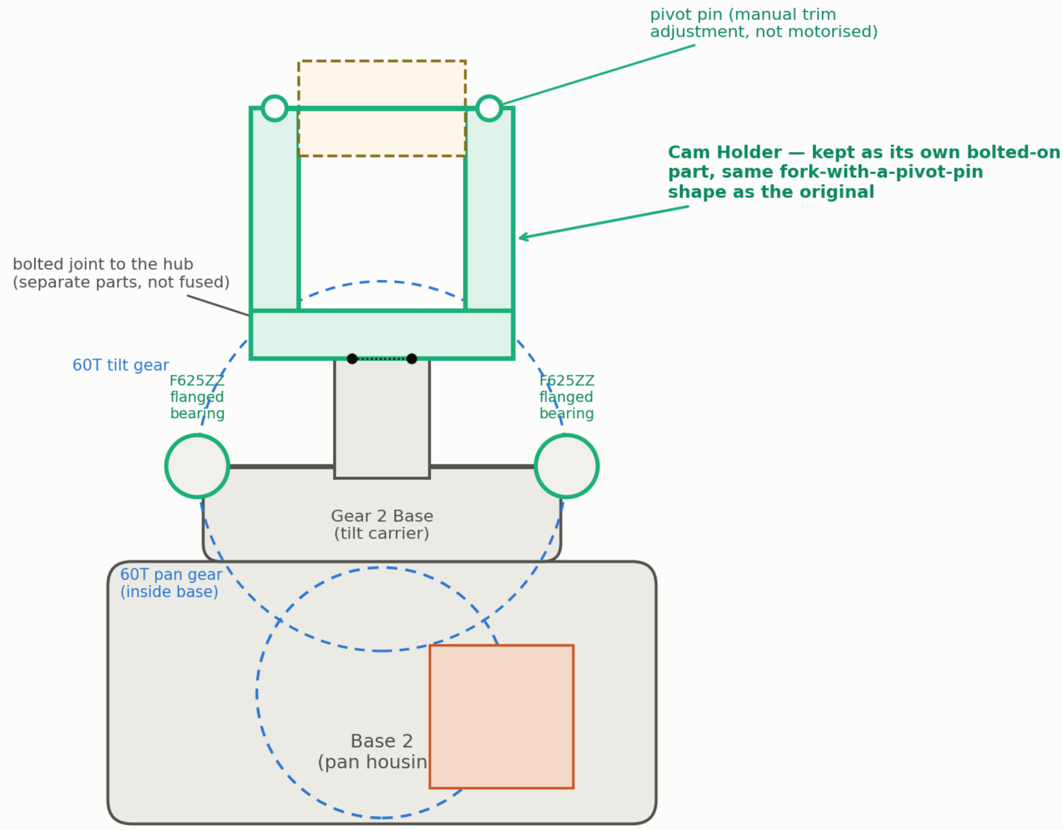
4.1 The “Cam Holder” stays its own part

In the original file, “Cam Holder” is a fork-shaped bracket — two arms with a hole through both of them on a shared line, like a hinge, holding whatever sits between the arms on a pin. It's also, by volume, the **second-largest single part in the entire original assembly** (only the main housing is bigger) — this was never a minor accessory, it was a major, deliberate structural piece.

There's a genuine open question about this part: its pivot hole doesn't line up with the motor-driven tilt axis — it's at right angles to it, and offset. We can't tell, just from the 3D geometry, whether that was meant to be a real third motorised axis, or whether it's simply an artefact of how the reference file was put together (this is explained fully in the original engineering analysis, and it genuinely can't be resolved without the original designer's own notes). The first redesign attempt sidestepped this question by deleting it — it fused Cam Holder directly into the tilt platform, so there was only ever one clear pivot and no ambiguity left to resolve.

Option B keeps Cam Holder as its own separate part, bolted onto the tilt platform rather than fused into it, with the same two-arm, common-pin shape as the original. To make that buildable rather than just decorative, its pivot pin gets a real job: it becomes a simple, hand-adjustable, friction-locked pivot — a manual trim or levelling adjustment for whatever payload sits in the fork, independent of the motor-driven tilt axis. This keeps the part, keeps its shape, keeps the puzzle the original left unanswered genuinely unanswered (rather than quietly deleting it) — and gives it something real to do in a design you can actually build.

Option B general arrangement, front view (schematic, not a manufacturing drawing)



PAN axis (vertical) — overall width shown to approximate scale

Figure 2 — Front view schematic. Cam Holder (green) bolts onto the tilt hub as its own part, rather than being fused into the platform beneath it. Its pivot pin runs through both arms, exactly as in the original.

4.2 The bearings keep their original "family," just downsized

A bearing is what lets a shaft spin smoothly inside a hole, instead of just wearing a groove into the plastic or metal around it. The original design used **flanged** bearings — bearings built with a small lip around the outside, with bolt holes in that lip, so the bearing bolts flat against a wall or plate. The first redesign attempt swapped these for a different, cheaper bearing family that just gets pressed into a plain round hole and held there by friction alone.

That swap was a reasonable engineering shortcut — but it wasn't forced. Small flanged bearings in the exact same core sizes the first attempt already chose (the well-known "608" and "625" bearing families) are just as easy to buy, cost about the same, and keep the "bolts flat to a face" character of the original. Option B uses those instead.

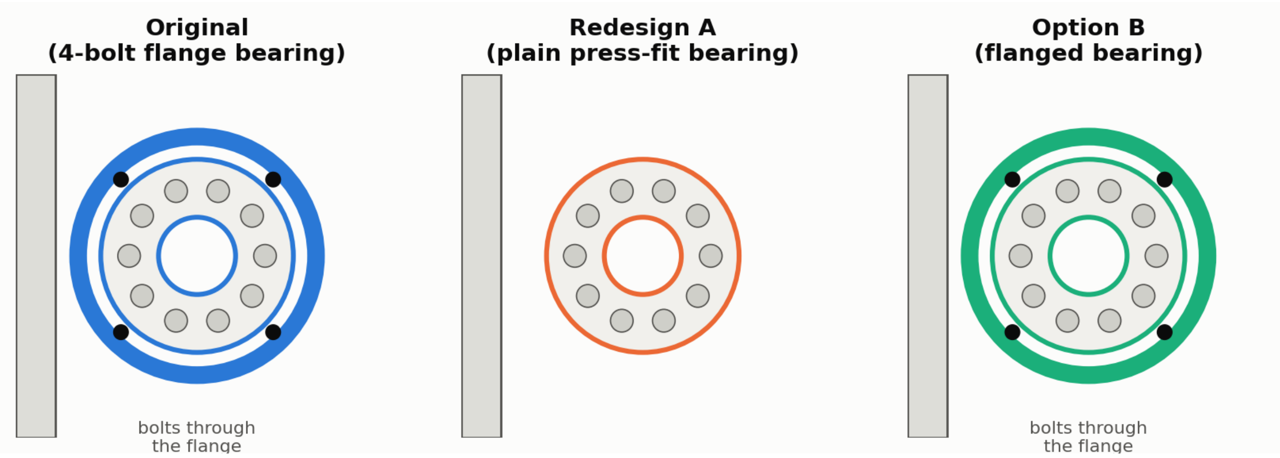


Figure 3 — The original bolted its bearings flat to a wall through a ring of small bolt holes. The first redesign switched to a plain press-fit bearing (still a perfectly good bearing, just a different family). Option B keeps the bolt-on flange, just at the smaller bore size the real shafts now need.

Pan shaft (bottom, main support)	Four-bolt flange bearing, 25 mm bore (this exact part had a broken 3D shape in the file — see section 6)	Plain press-fit bearing, 8 mm bore	Flanged bearing, 8 mm bore, bolted to the housing
Tilt shaft (both ends)	"Wall" bearings, 37.25 mm bore, x2	Plain press-fit bearings, 5 mm bore, x2	Flanged bearings, 5 mm bore, x2, bolted flat as in the original

Why the bore sizes had to shrink: at 306 mm across, the original's shafts were 25 mm and 37 mm thick — there's no version of those bearings small enough to sell for an 8 mm or 5 mm shaft, so a straight scale-down is impossible. That part of the change is forced by real hardware catalogues, not by taste. Which specific bolt-on bearing family to use, though, was a free choice — and here it's chosen to match the original's "flange" character rather than the cheapest possible option.

4.3 The little spacer parts stay separate parts

The original used four small identical spacer sleeves — short printed tubes that sit on a shaft between a gear and a wall, just to hold things the right distance apart. The first redesign deleted these as separate parts and turned them into little raised bumps printed directly onto the neighbouring parts instead — fewer things to lose, but also fewer things you can replace individually if one wears out or you need a different thickness. Option B keeps them as their own small, separate, resized parts, exactly as the original modelled them.

4.4 The motor mounting plate stays a separate, bolt-on part

The original modelled a distinct, thin mounting plate (“Motor base”) that a motor bolts to, rather than printing the motor pocket directly into the main housing. Option B keeps that separation for the pan motor: if a motor ever needs replacing, or you want to test a different one, you unbolt one small plate instead of reprinting the entire base.

5. What still had to change, and why

Keeping the original's spirit doesn't mean nothing changes — some things are genuinely forced, either by the real motor's shape or by what's actually possible to 3D print reliably at this smaller size. Option B makes exactly the same calls the first attempt made here, because these particular calls were sound engineering, not stylistic simplification.

5.1 Gear tooth size shrinks slightly

Gears are described by a “module” number, which is really just a standard way of saying how big each tooth is — a bigger module means bigger, chunkier teeth. The original used module 1.125. At the new, smaller size, Option B uses module 1.0 instead (a roughly 11% reduction in tooth size), while keeping the exact same tooth counts (20 teeth on the small gear, 60 on the big one) and the same 3-to-1 ratio.

This isn't a style choice — it's about what a 3D printer's nozzle can actually lay down cleanly. At module 1.0, each tooth is about four nozzle-widths thick at a common 0.4 mm nozzle size; drop the module any further and each tooth becomes only two to three nozzle-widths thick, at which point how a tooth actually prints starts depending more on the printer's software guesses than on the drawing — and the gap between meshing teeth (“backlash”, explained in the glossary) stops being predictable. The teeth are nowhere near their strength limit at this size, so module 1.0 is chosen purely for printing reliability.

5.2 Two small bearings on the motor shafts are dropped

The original had two tiny bearings sitting on each motor's output shaft, just past the motor body, to support the shaft before it reaches the small gear. The real 28BYJ-48 doesn't need this: its shaft is already held steady inside its own internal gearbox, and only pokes out a few millimetres before the gear sits on it — nowhere near far enough to need extra support. Adding a bearing here would be extra cost and assembly complexity solving a problem the real motor doesn't have. This is dropped in both designs, and it's a case where the earlier redesign's judgement was already right.

5.3 The motor itself becomes a real, connected part

One quirk of the original file is worth mentioning: it did include a part explicitly named “Stepper motor 28BYJ-48 5V” — but that part was floating off to one side, not actually touching or connected to either gear train. Both redesigns fix this by making the real motor the thing that actually drives the gears, in the position it needs to be in to do so. This is a genuine correction, not a simplification.

6. Fixing what was actually broken in the original

Separately from resizing anything, the original reference file has a handful of real problems — not questions of taste, but places where the 3D geometry genuinely doesn't work. Your instruction was clear that fixing real, provable problems like these is fine and expected; it's not the same thing as “simplifying.” Option B fixes these the same way the first attempt did, because the fix is correct regardless of which specific bearings or brackets you choose around it.

6.1 One bearing part had an invalid 3D shape

The original's four-bolt flange bearing — the one supporting the main, pan-axis shaft — doesn't just have a tight fit; part of its own 3D shape fails a basic soundness check (the kind of check that answers “is this actually a solid, physically-possible object, or does the geometry contradict itself somewhere?”). A shape like this can look fine on screen but would confuse a 3D printer's slicing software or a manufacturing process. Since this exact bearing size doesn't exist at Option B's smaller scale anyway (section 4.2), replacing it with a real, off-the-shelf bearing fixes this automatically — there's no version of the new part with the same flaw.

6.2 Four places where two solid parts were told to occupy the same space

A “clash” or “interference” means two solid parts' 3D shapes genuinely overlap each other in space — something that's fine in a drawing but physically impossible to build, like two people trying to stand in the exact same spot. The original reference file has four real clashes like this, all connected to how the bearings sit against their shafts and housings — not because the bearings were the wrong size (the sizes were actually correct), but because in this particular saved snapshot of the file, the parts weren't lined up dead-centre with each other, and nowhere was material actually cut away to make room for the bearings' outer edges.

Option B avoids all four by following one simple rule everywhere a bearing sits in a pocket: **the housing only ever touches the bearing's outer ring, and the shaft (and anything else on the shaft) only ever touches the bearing's inner ring — never both from the same part.** Every inside corner where a shape meets a shape also gets a small relief cut, because a 3D printer naturally leaves a rounded fillet in every internal corner, and an unrelieved corner would push the bearing slightly out of square. The practical build tip that goes with this: print a small test block with a few slightly different hole sizes, press-fit a real bearing into each, and use whichever size gives a firm (not loose, not forced) fit for every bearing pocket in the design — 3D printers consistently print round holes a little undersized, by an amount that varies printer to printer.

7. Full part list

16 different part designs, 22 individual parts once you count the ones used more than once — compared to the original's 17 designs / 25 parts, and the first attempt's 11 designs / 19 parts. Option B keeps nearly as many distinct part designs as the original itself.

1	Base 2 (pan housing)	1	Print	Kept — resized, redrawn pan-motor recess and bearing pocket	Houses the pan gear, the flanged pan bearing, and the pan motor
2	Motor base (pan motor plate)	1	Print	Kept — resized for the real motor	Bolts to Base 2; separate for easy motor swaps
3	Gear 2 Base (tilt carrier)	1	Print	Kept — resized, new bearing pockets	Sits on the pan shaft; carries the tilt gear and both tilt bearings
4	Baring 2 shaft (tilt hub)	1	Print	Kept — resized	Keys the tilt gear onto the tilt shaft; Cam Holder bolts to this
5	Cam Holder	1	Print	Kept — resized, same fork-and-pivot shape, now a manual trim mount	See section 4.1
6	Tilt motor bracket	1	Print	New — forced by the real motor's offset shaft	Holds the tilt motor ~40 mm behind the tilt axis
7	Spur gear, 60 teeth (pan)	1	Print	Adapted — module 1.125 → 1.0, 8 mm bore	Module change explained in section 5.1
8	Spur gear, 60 teeth (tilt)	1	Print	Adapted — module 1.125 → 1.0, 5 mm bore	Same as #7, different bore for the tilt shaft
9	Spur gear, 20 teeth (pinion)	2	Print	Adapted — module change, flat-sided bore for the real motor shaft	Identical part on both motors — both use the same shaft
10	Spacer sleeve	4	Print	Kept — its own small part, resized	Not merged into bosses on other parts (section 4.3)
11	Pan shaft	1	Cut steel rod, 8 mm	Adapted — real metal rod, resized from the original's 25 mm placeholder	Printed shafts wear oval over time; metal is used here as in the first attempt
12	Tilt shaft	1	Cut steel rod, 5 mm	Adapted — real metal rod, resized from the original's 37 mm placeholder	
13	Cam Holder pivot pin	1	Cut steel rod, 3 mm	New — the original didn't model a separate pin for this pivot	Needed to physically build the two-arm fork joint at this scale
14	28BYJ-48 stepper motor + ULN2003 driver board	2	Buy	Fixes the original's disconnected placeholder motor	One per axis; see section 3
15	Flanged bearing, 8 mm bore (608-family)	1	Buy	Substituted — forced (original's 25 mm bore doesn't exist this small); kept the bolt-on flange	Pan shaft, base support

16	Flanged bearing, 5 mm bore (625-family)	2	Buy	Substituted — forced (original's 37 mm bore doesn't exist this small); kept the bolt-on flange	Tilt shaft, both ends, ~16 mm apart

Dropped from the original, and why: the two tiny motor-shaft bearings (the real motor doesn't need them, section 5.2), the two generic placeholder “motor body” parts and the two placeholder “motor shaft” parts (replaced entirely by the real, bought 28BYJ-48 motors, which already include their own body and shaft).

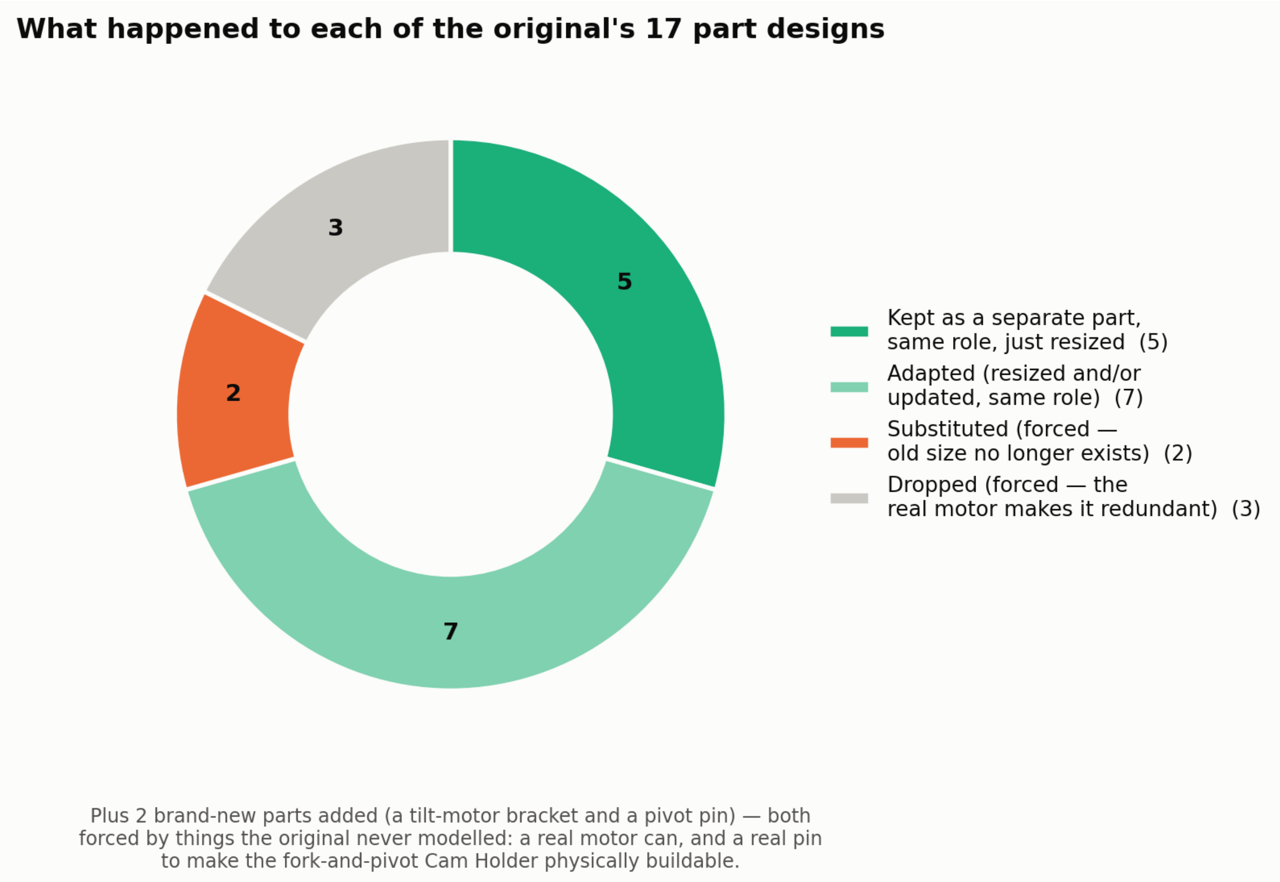


Figure 4 — What happened to each of the original's 17 part designs. Most were kept or lightly adapted; only two were substituted (forced by unavailable bearing sizes) and three dropped (made redundant by the real motor). Two brand-new parts were added, both forced by the real motor or the smaller build scale.

8. Size, in numbers

Overall size (width x height x depth)	306.5 x 298.1 x 303.7 mm	110 x 126 x 85 mm	**~120 x 140 x 92 mm**
Box volume (a rough "how much space it takes up")	27,748 cm^3	1,178 cm^3	**~1,546 cm^3**
Shrink vs. the original (linear)	—	2.8x smaller	**~2.6x smaller**
Shrink vs. the original (by volume)	—	24x smaller	**~18x smaller**
Different part designs / total parts	17 / 25	11 / 19	**16 / 22**
Gear tooth size ("module")	1.125	1.0	**1.0**
Reduction on each axis	3-to-1 (motor unknown)	3-to-1, plus ~64-to-1 inside the motor	**Same as first attempt**
The Cam Holder question	Unresolved — 2 or 3 rotation axes, unclear	Resolved by deletion — folded into the tilt platform	**Resolved by keeping it — a real, separate, hand-adjustable third pivot**
Physically-impossible overlaps between parts	4 real clashes, ~1,417 mm^3	0, by design	**0, by design (same fix, section 6.2)**

A note on precision: the Option B figures above are a reasoned layout estimate, built from the same real building blocks the first attempt used (the motor's actual can size and shaft position, the gears' actual tooth geometry, the bearings' actual catalogue dimensions) plus a small, consistent allowance for every bolted joint that a fused design wouldn't need. They are not measurements taken from a finished 3D model — no new CAD file was built for this round, by design, since this is a written design study. Treat these numbers the same way the first attempt's own document asked you to treat its numbers: a solid design target to build a real 3D model against, not a manufacturing drawing to cut metal from directly.

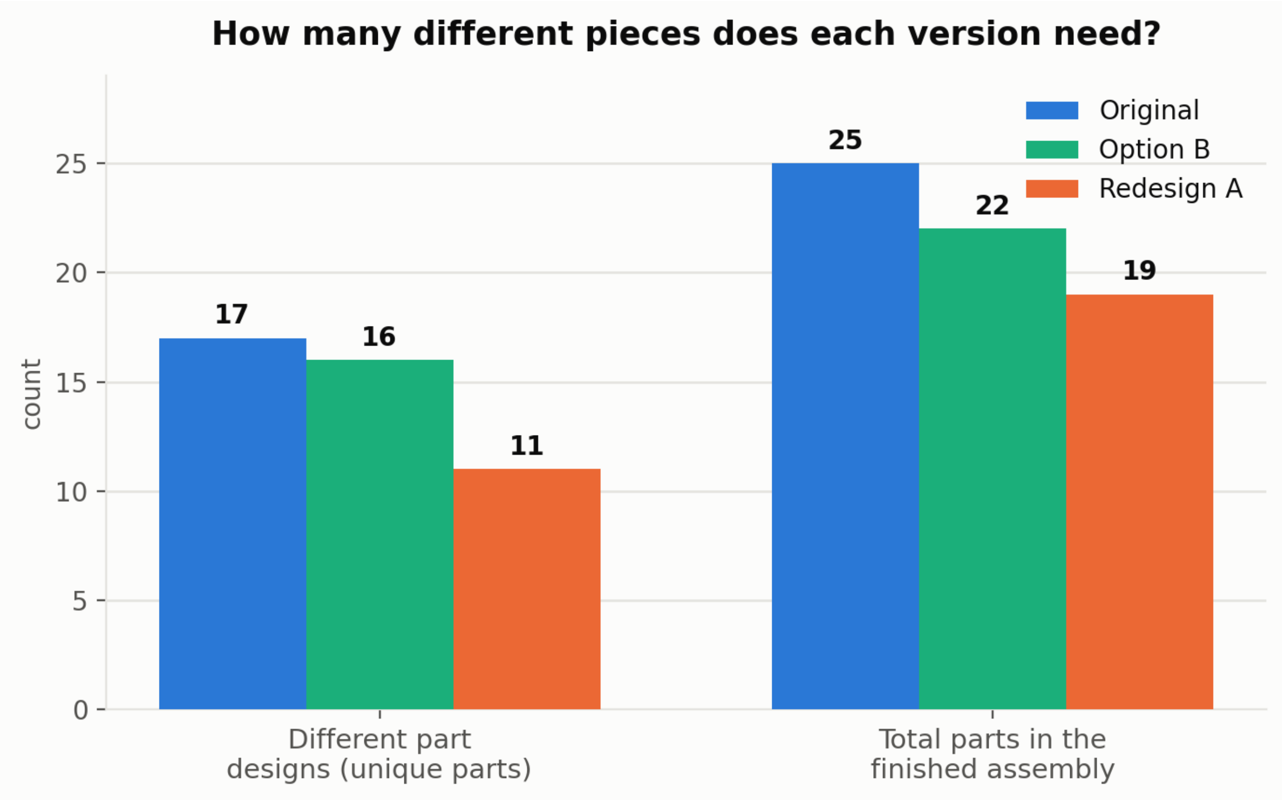


Figure 5 — Option B keeps nearly as many distinct part designs as the original (16 of 17), while the first attempt cut that number roughly in half (down to 11).

9. General arrangement

These are layout schematics — they show how the pieces relate to each other and roughly where things sit, not a manufacturing drawing. Component outlines are simplified rectangles and circles standing in for the real part shapes.

Option B general arrangement, front view (schematic, not a manufacturing drawing)

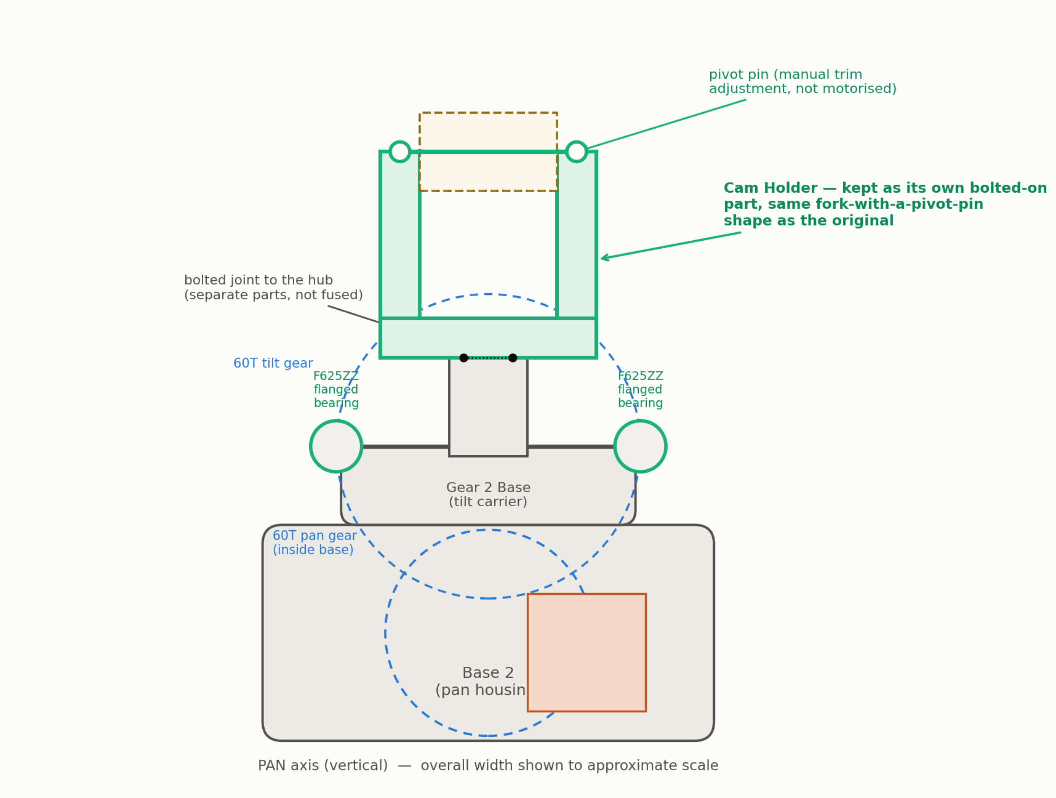


Figure 6 — Front view, looking along the pan axis.

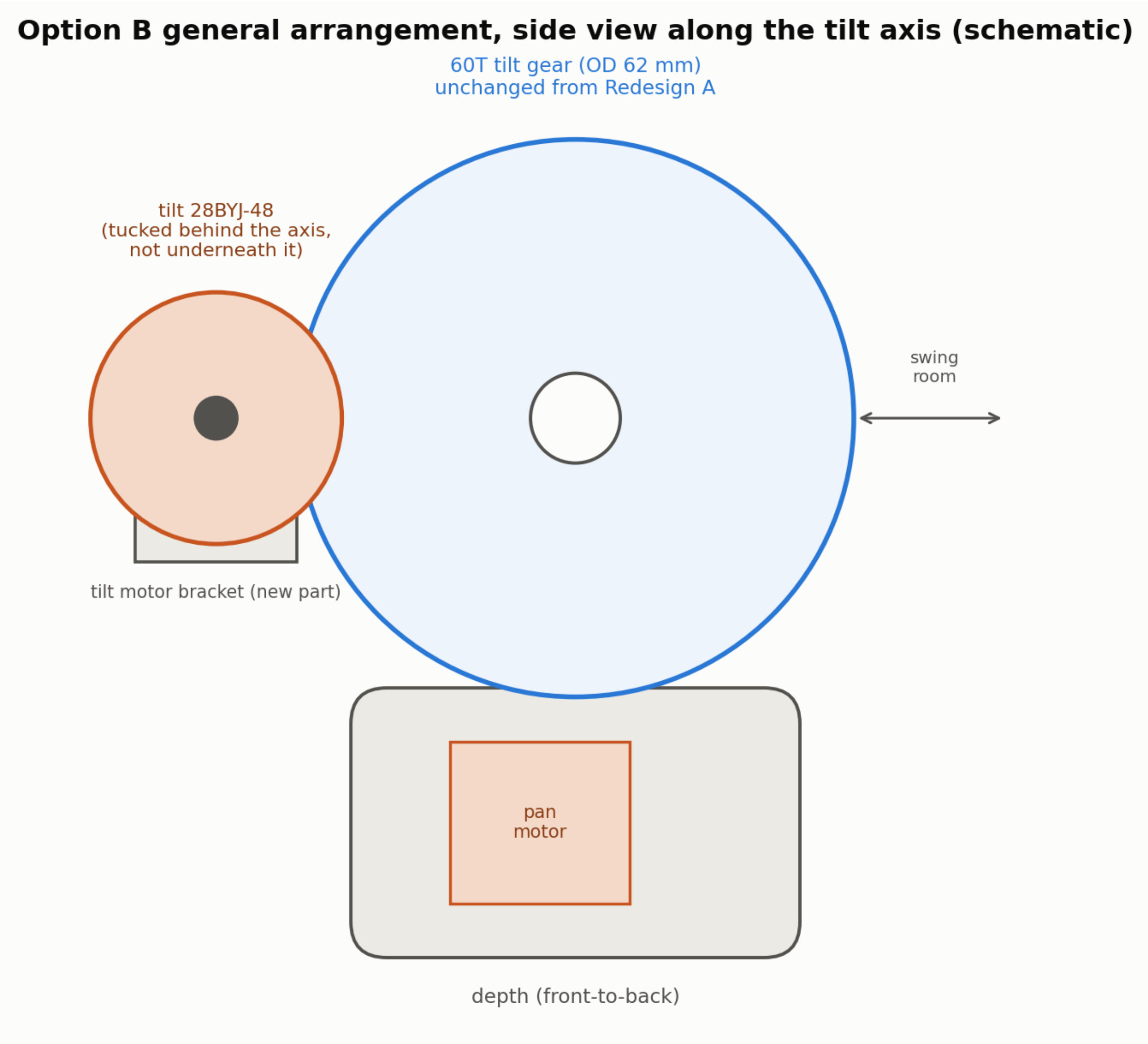


Figure 7 — Side view, looking along the tilt axis. Depth (front-to-back) is the cheapest dimension to spend here, which is why the tilt motor tucks in behind the gear rather than sitting underneath it — putting it underneath would make the whole head taller instead.

10. Does the motor actually have enough push? (strength check)

Because Option B uses the exact same motor, the exact same gear ratio, and the exact same tooth module as the first redesign attempt — the only things that changed are the surrounding brackets and bearings — this strength check carries over unchanged. Nothing about how the mechanism is driven is any different.

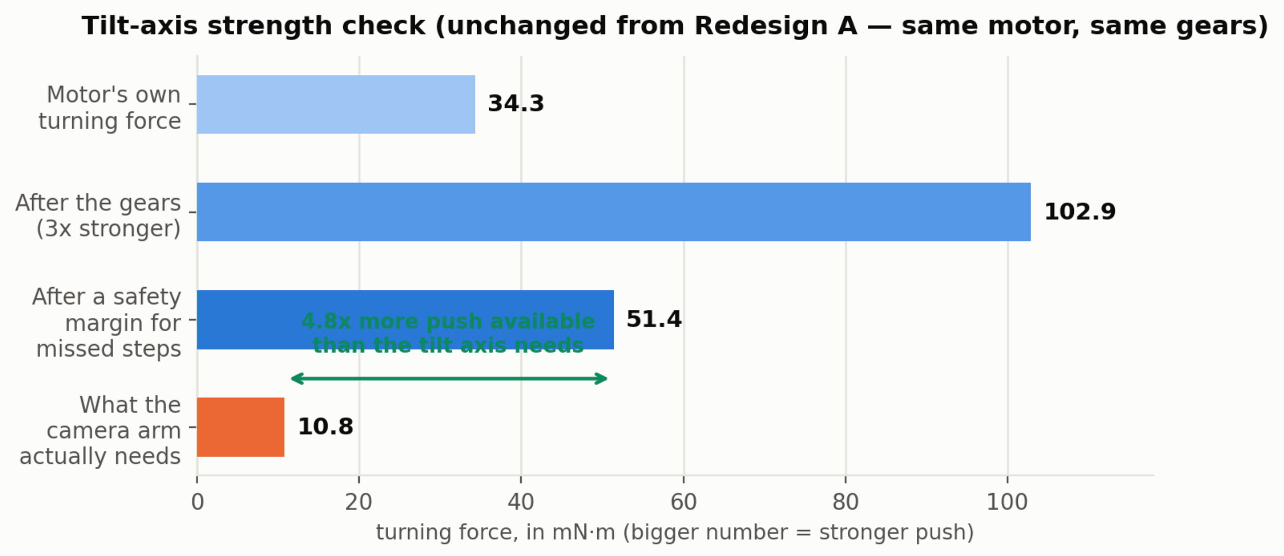


Figure 8 — Reading this bottom-to-top: the motor's own turning force, multiplied by the gears, gives you the total push available. A safety margin is set aside for occasionally missed motor steps. What's left is still more than 4.5 times what a small 15-gram payload actually needs to hold and move against gravity on the tilt axis. The pan axis needs even less — spinning side-to-side doesn't fight gravity at all, only friction.

One thing to keep in mind because Option B's Cam Holder sits as its own bolted-on part rather than fused flush against the tilt platform: the payload may end up sitting slightly further out from the tilt axis than the first attempt assumed (roughly 50 mm instead of 40 mm, as a conservative estimate). Even at that longer reach, there's still very comfortably enough spare turning force — about 3.8 times what's needed instead of 4.8 times. Confirm the real arm length once a 3D model exists, but this is not expected to be close to a problem.

Holding torque

Holding position with the power off: multiply the motor's own internal gearing (about 64-to-1) by the external gears (3-to-1) and you get roughly 190-to-1 overall. Combined with the motor's own resistance to being back-driven, a 15-gram payload will simply sit where you leave it when the power is switched off — no brake, no counterweight, and no separate locking mechanism needed. This is unchanged from the first attempt, and applies equally to Option B.

11. What to check before you build this

■ **Measure your actual motor.** The shaft offset and can size used throughout this document (and the first redesign attempt before it) come from a typical datasheet for this widely-cloned motor. Real dimensions vary a little between suppliers — measure the specific motor you actually have with calipers before finalising the bracket dimensions in CAD.

■ **The Cam Holder question is still genuinely open.** Option B resolves the original's ambiguous third pivot by making it a real, deliberate, hand-adjustable feature — but that's a design decision made without the original designer's own notes. If the original was actually meant to have a real, motorised third axis (not a manual trim feature), Option B's version won't provide powered motion there — only manual adjustment. Worth a sanity check with whoever built the original reference model, if that's possible.

■ **The motor's true gearing isn't exactly 64-to-1.** This particular motor is widely reported to actually gear down at about 63.68-to-1, not a clean 64-to-1 — a small, roughly half-a-percent error that doesn't matter for small movements but will slowly drift if you try to track an exact position purely by counting motor steps over many turns. Adding a simple home switch on each axis (a small detector that says “you are now at zero”) sidesteps this cleanly.

■ **Expect a small amount of play (“backlash”) between the gear teeth.** A little looseness between printed gear teeth is normal, roughly ten times bigger than the motor's own smallest step size. This doesn't hurt accuracy as long as you always approach a target position from the same direction — worth designing the control software around from the start.

■ **Recheck clearances once a real 3D model exists.** This document's numbers are a careful layout estimate, not measurements taken from finished geometry (see the note in section 8). Before printing anything, build the actual 3D model and check tight spots — the original had a genuinely tight 3 mm clearance between the tilt gear and the platform beneath it, and a similar check should be repeated here.

■ **Consider the 12V version of the same motor.** Same physical size, same mounting, but roughly double the turning force, for free — worth it if your electronics can supply 12V instead of 5V, which would roughly double every safety margin quoted in section 10.

12. Plain-English glossary

A few terms used above, explained without assuming any engineering background.

Bearing
A small ring-shaped part that lets a shaft spin smoothly inside a hole, instead of grinding directly against the material around it and wearing a groove into it.

Flanged bearing
A bearing built with a small lip around its outside edge, with tiny bolt holes in that lip, so it can be bolted flat against a wall or plate rather than just pressed into a round hole.

Gear module
A standard number describing how big each individual gear tooth is. A bigger module means bigger, chunkier teeth; a smaller module means finer, smaller teeth.

Pressure angle
A standard angle (usually 20 degrees) that defines the exact curved shape of a gear tooth's flank. As long as two meshing gears share the same pressure angle and tooth size, they'll mesh correctly — it's a manufacturing standard, not something you need to design yourself.

D-flat / double-D shaft
A normally round shaft with one or two flat sides ground onto it, like a two-sided key. A gear or wheel with a matching flat-sided hole can't spin loose on the shaft the way it would on a perfectly round one.

Backlash
The small amount of play or looseness between two meshing gear teeth — similar to the bit of give you feel in a car's steering wheel before the front wheels actually start to turn.

Holding torque
How much outside force it would take to overpower a motor and move it by hand while the power is switched off. High holding torque means the mechanism stays put on its own, with no separate brake needed.

Resolution (degrees per step)
The smallest angle a motorised axis moves with each individual electrical pulse sent to the motor. A smaller number means finer, more precise aiming.

Interference / clash
When two solid 3D parts' shapes genuinely overlap in space — physically impossible to build, even if it looks fine as a picture on a screen, like two people trying to stand in the exact same spot.

Fit / clearance
How much (or how little) intentional gap is designed between two parts that touch — a shaft and the hole it spins in, for example. Too little gap and the parts bind or won't go together at all; too much and things wobble.

Valid 3D shape (“topology”)
A basic soundness check on a 3D model: does this shape actually describe a real, physically-possible solid object, with no internal contradictions? A shape can look correct on screen and still fail this check, which then causes real problems for 3D printing or manufacturing software downstream.

End of report. This is a layout study for design and discussion purposes — treat every dimension as a well-reasoned starting point for a real 3D model, not as a finished manufacturing drawing.